Utility/Privacy Trade-off through the lens of Optimal Transport

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Different notions of privacy

- **Differential Privacy:** cannot retrieve info from databases.
- **Economic Privacy:** info belief unchanged (Bayesian).

$\rightarrow$ long-term utility. Info is a resource.
Find best privacy/revenue trade-off?

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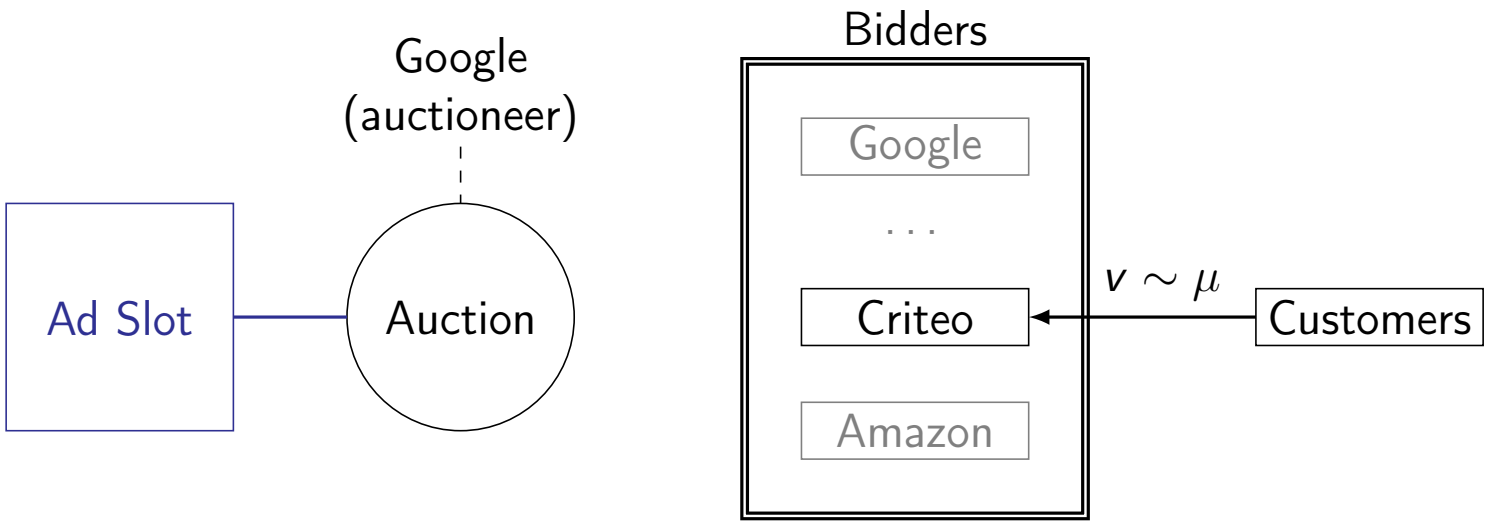
Motivation: repeated auctions

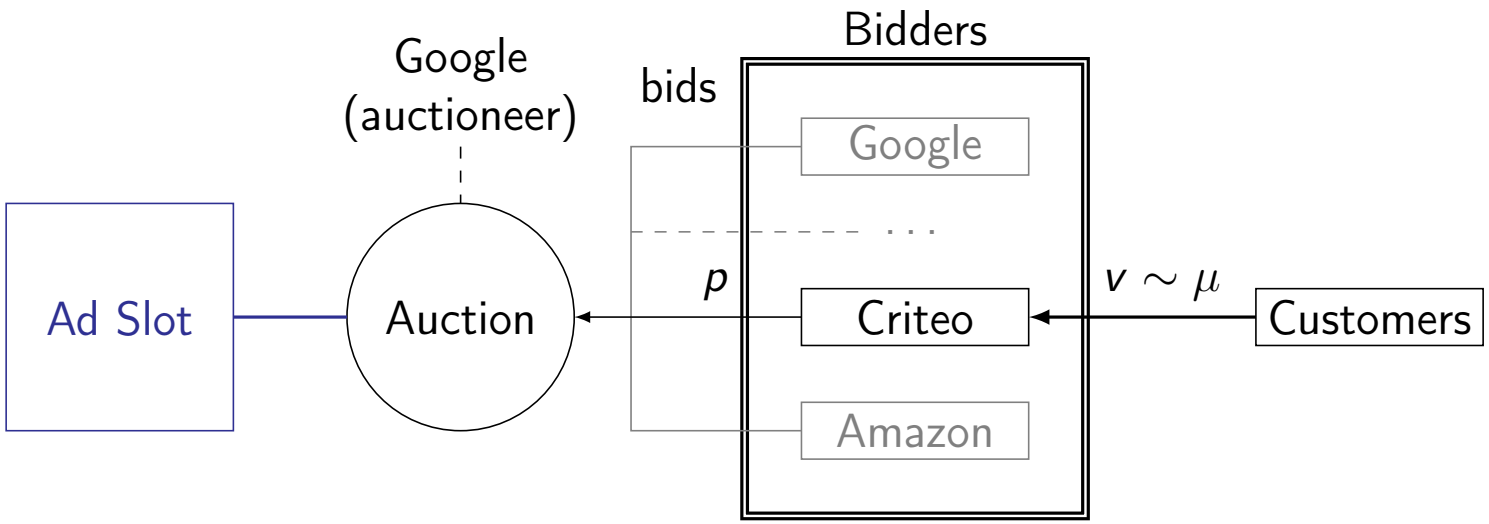
Ad slot valued $v \sim \mu$.



Online advertisement auction system.

Motivation: repeated auctions

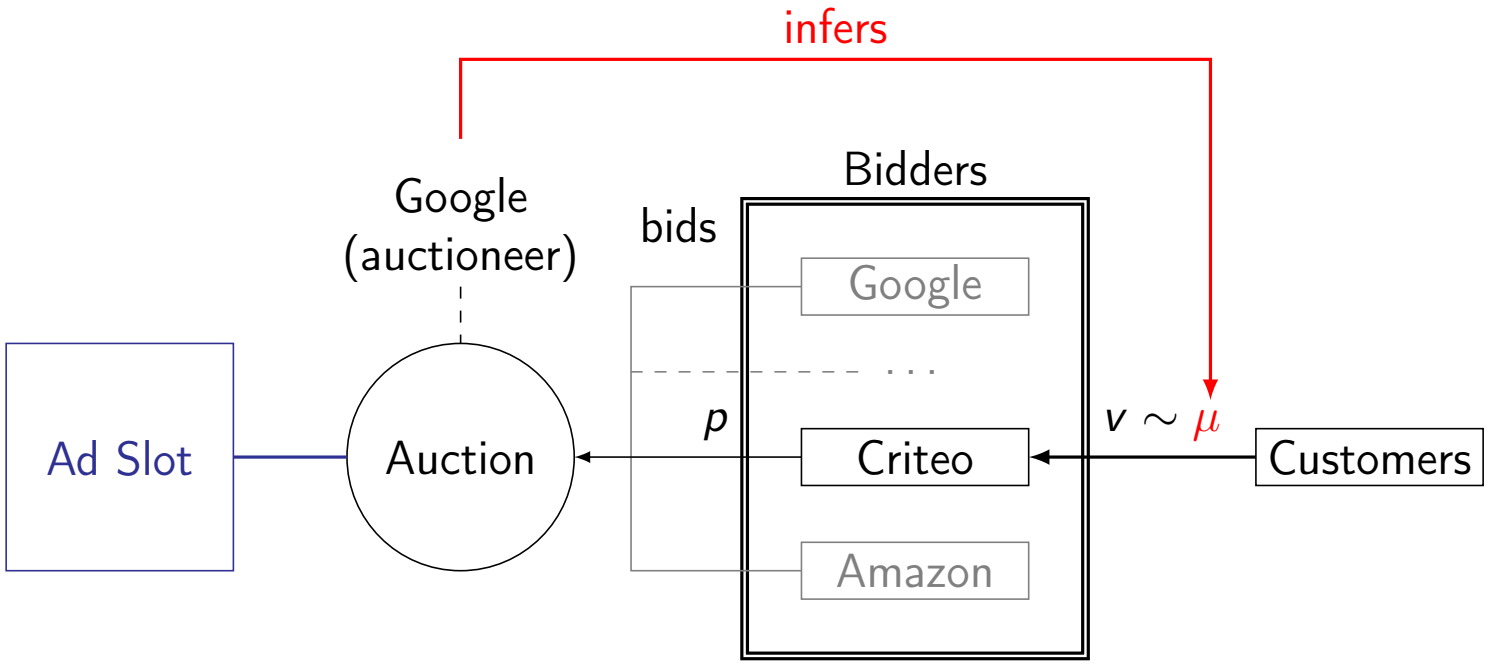
Ad slot valued $v \sim \mu$. Bid p



Online advertisement auction system.

Motivation: repeated auctions

Ad slot valued $v \sim \mu$. Bid $p \implies$ auctioneer infers μ.



Online advertisement auction system.

If μ public, auctioneer's revenue $\nearrow$ bidder's utility $\searrow$
Bidder's goal: short term utility **and** hide μ

Toy example

Player: type $y_k \sim p_0$ with $k \in \{1, \ldots, K\}$
action $x \in \mathcal{X}$, loss $x^\top y_k$

$$\min_{x \in \mathcal{X} \subset \mathbb{R}^d} x^\top y_k$$

Adversary: observes x and infers k

If $x|k \sim \mu_k \in \mathcal{P}(\mathcal{X}) \to$ posterior p_x given by Bayes rule:

$$p_x(k) = \frac{p_0(k)\mu_k(x)}{\sum_{j=1}^K p_0(j)\mu_j(x)}$$

Toy Example

Optimization Program

Existing approach: constrained optim[1]

$$\min_{\mu_1,\dots,\mu_K} \sum_{k=1}^{K} p_0(k)\, \mathbb{E}_{x\sim\mu_k}\left[x^\top y_k\right] \quad \text{s.t.} \quad \mathbb{E}[KL(p_x, p_0)] \leq \varepsilon$$

Our approach: Lagrange form + actual trade-off

$$\boxed{\min_{\mu_1,\dots,\mu_K} \sum_{k=1}^{K} p_0(k)\, \mathbb{E}_{x\sim\mu_k}\left[x^\top y_k + \lambda\, KL(p_x, p_0)\right]}$$

[1] Eilat, R., Eliaz, K., and Mu, X. (2019). Optimal privacy-constrained mechanisms

General formulation

$$\inf_{\substack{\gamma \\ \pi_2 \# \gamma = p_0}} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) + \lambda \, D(p_x, p_0) \, \mathrm{d}\gamma(x, y) \qquad \text{(PRP)}$$

- private type $y \sim p_0 \in \mathcal{P}(\mathcal{Y})$
- $\pi_2 \# \gamma$ marginal distrib. of γ on $\mathcal{Y}$
- Toy example: $c(x, y) = x^\top y$ and $D = KL$

Convexity

Definition (f-divergence)

D is an f-divergence if there is $f : \mathbb{R}_+^* \to \mathbb{R}$ convex, $f(1) = 0$ s.t. for all distributions P, Q: $D(P, Q) = \int f(\frac{\mathrm{d}P}{\mathrm{d}Q})\mathrm{d}Q$.

Examples: *KL, reverse KL and TV.*

Theorem (Convexity)

D is an f-divergence $\implies$ (PRP) *is convex in* γ.

Easy for finite action **and** type spaces

Finite prior support

Theorem

If $|\mathrm{supp}(p_0)| = K$*, there exists a solution of* (PRP) *with support size* $K(K+2)$*.*

(PRP) equivalent to

$$\boxed{\begin{array}{c} \displaystyle\inf_{\substack{\gamma \in \mathbb{R}_+^{(K+2)K} \\ x \in \mathcal{X}^{K+2}}} \sum_{i,k} \gamma_{i,k} c(x_i, y_k) + \lambda\, \mathbb{E}[\mathrm{D}(p_x, p_0)] \\ \text{s.t. } \forall j,\ \sum_i \gamma_{i,j} = p_0^j \end{array}} \qquad \text{(Disc-Opt)}$$

$\rightarrow$ finite dimension ☺ but not jointly convex ☹

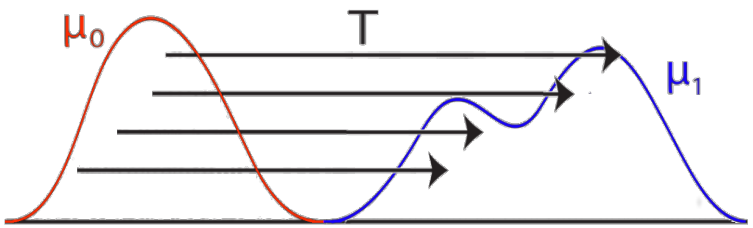
Optimal Transport

Optimal transport between μ and ν:

$$\boxed{\inf_{\gamma \in \Pi(\mu,\nu)} \int c(x,y)\,\mathrm{d}\gamma(x,y)}$$

with $\Pi(\mu,\nu) = \left\{\gamma \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}) \mid \pi_1 \# \gamma = \mu \text{ and } \pi_2 \# \gamma = \nu\right\}$

- compare distributions
- widely used in recent applications



Sinkhorn divergence

Drawback of OT: heavy computational cost

Definition (Sinkhorn divergence)

$$\mathrm{S}(\mu,\nu) = \inf_{\gamma \in \Pi(\mu,\nu)} \int c \, \mathrm{d}\gamma + \lambda \int \log\left(\frac{\mathrm{d}\gamma}{\mathrm{d}\mu \mathrm{d}\nu}\right) \mathrm{d}\gamma$$

Entropic regularization $\implies$ fast OT distances approximation[2]

[2] Cuturi, M. (2013). Sinkhorn distances: Lightspeed computation of optimal transport

Sinkhorn divergence minimization

Theorem

If D=KL, (PRP) *equivalent to*

$$\inf_{\mu \in \mathcal{P}(\mathcal{X})} \mathrm{S}(\mu, p_0) \tag{1}$$

Proof.

$$\inf_{\mu \in \mathcal{P}(\mathcal{X})} \inf_{\gamma \in \Pi(\mu, p_0)} \int c \mathrm{d}\gamma + \lambda \int \log\left(\frac{\mathrm{d}\gamma}{\mathrm{d}\mu \mathrm{d}p_0}\right) \mathrm{d}\gamma$$

Sinkhorn divergence minimization

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Bayes rule: $\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\mu(x)} = \mathrm{d}p_x(y)$.

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$$\underbrace{\inf_{\mu \in \mathcal{P}(\mathcal{X})} \inf_{\gamma \in \Pi(\mu, p_0)}}_{\inf\limits_{\substack{\gamma \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}) \\ \pi_{2\#}\gamma = p_0}}} \int c \mathrm{d}\gamma + \lambda \underbrace{\int \log\left(\frac{\mathrm{d}\gamma}{\mathrm{d}\mu \mathrm{d}p_0}\right) \mathrm{d}\gamma}_{= KL(p_x, p_0)}$$

■

Expe 1: Toy example

Disc: descent on (Disc-Opt)
Sink: descent on (Sink-Opt)
DC: adapted to linear loss

$$\min_{\substack{(\gamma,x)\in\Delta_{(K+2)K}\times\mathcal{X}^{K+2}\\ \pi_{2\#}\gamma=p_0}} \sum_{i,k}\gamma_{i,k}c(x_i,y_k)+\lambda\mathbb{E}[\mathrm{KL}(p_x,p_0)] \qquad \text{(Disc-Opt)}$$

$$\min_{(\alpha,x)\in\Delta_{K+2}\times\mathcal{X}^{K+2}} \mathrm{S}\Big(\sum_i \alpha_i\delta_{x_i},p_0\Big) \qquad \text{(Sink-Opt)}$$

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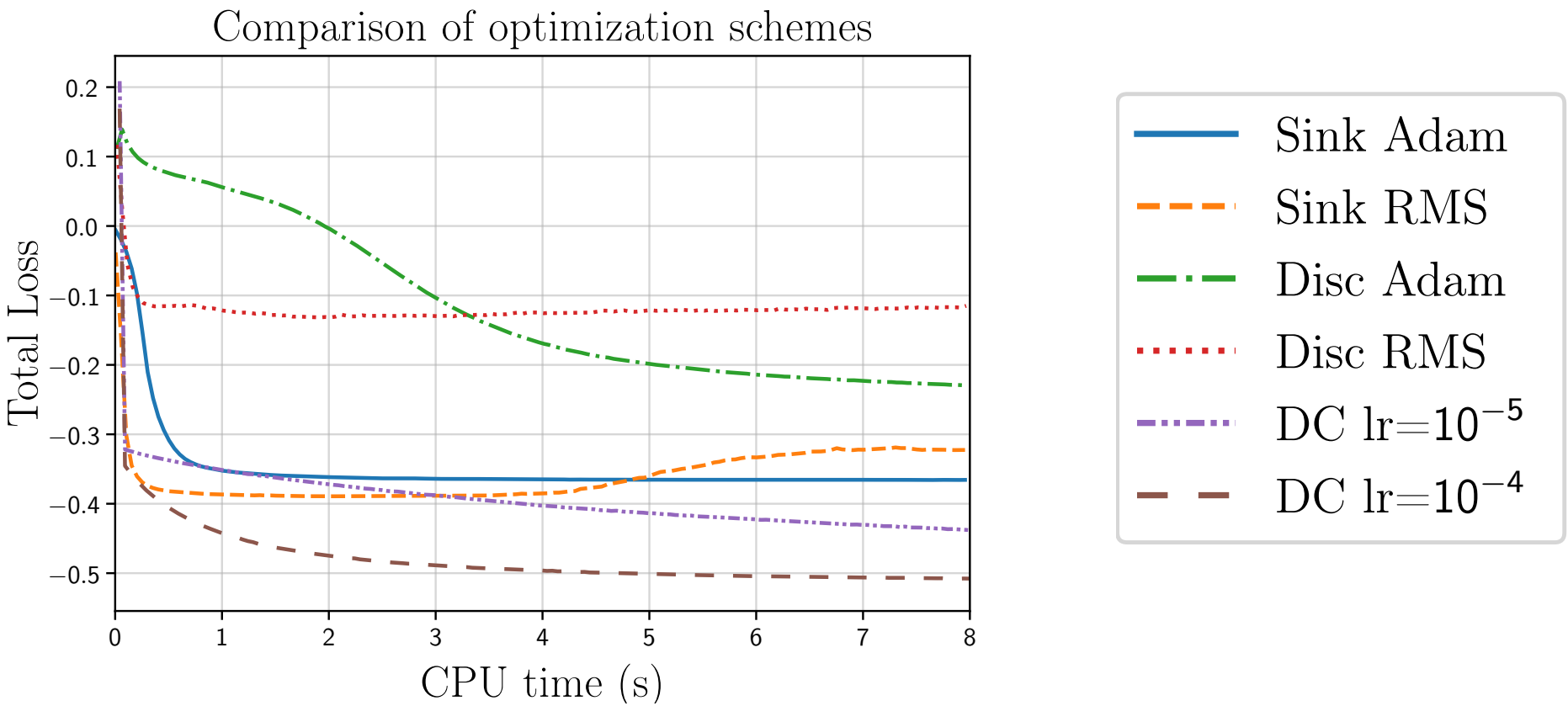


DC $\succ$ Sink $\succ$ Disc $\rightarrow$ better understanding of problem structure

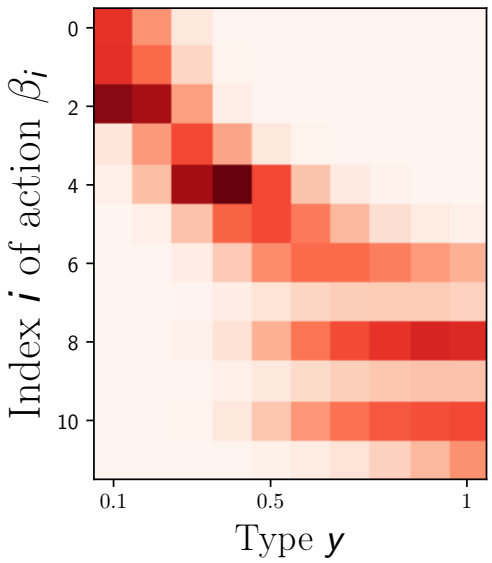
Expe 2: repeated auctions

Problem description

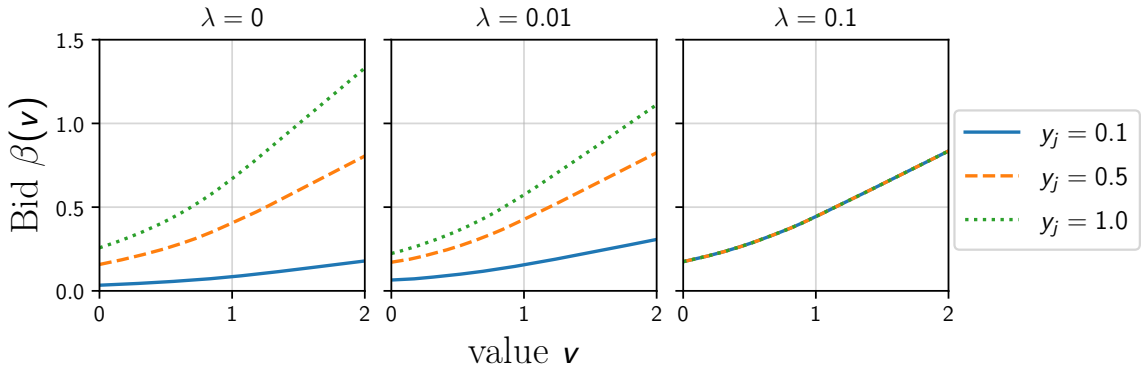
- **Private type:** value distribution $\mu_{y_j} = \mathrm{Exp}(\frac{1}{y_j})$ with $y_j \in (0, 1]$
- **Action:** mapping $\beta_i : \mathbb{R}_+ \to \mathbb{R}_+$ (parametrized by NN)
- action i and type j: bid function $\beta_i(v/y_j)$
- Known cost function[3]

[3] Nedelec, T., Karoui, N. E., and Perchet, V. (2019). Learning to bid in revenue-maximizing auctions

Expe 2: repeated auctions



(a) Heat-map of joint distribution γ ($\lambda = 0.01$)



(b) Evolution of the most used β_i with type and regularization constant

- Randomize over close mappings
- Type is hidden (revealed) when close to 1 (0) $\rightarrow$ heterogeneous value of information
- from optimal ($\lambda = 0$) to non-revealing strategies ($\lambda = 0.1$)

Recap

- utility/privacy trade-off motivated by economic mechanisms
- general regularized formulation
- equivalent to Sinkhorn divergence minimization
- computationally tractable schemes

Future work:

- optimization schemes for Sinkhorn minimization
- estimate Pareto frontier?

Slides, code and paper at `eboursier.github.io`

Thank you!